Graph Algorithms Lab: BFS, DFS, Dijkstra

LAB INSTRUCTION | any programming language | black and white print version

Tashkent Webster University | Data Structures 2 | Fall 2026

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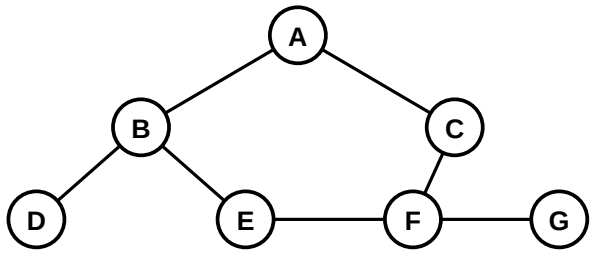
Goal

Implement three classic graph algorithms in **any programming language** you know (C, C++, Java, Python, C#, JavaScript, ...). You will do **3 tasks**: one for BFS, one for DFS and one for Dijkstra. Each task has a hand trace on paper, a program, and short written answers.

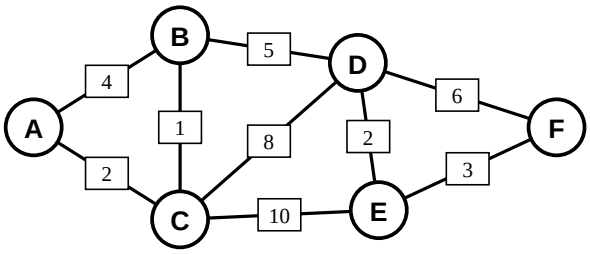
Rules

- Build the queue and the stack yourself (an array, list or linked list). Do not use a ready-made graph library.
- The graph data is given below. You may type it directly into your program.
- Examine neighbours in **alphabetical order** (A before B). Your output must match the expected output **exactly**, line by line.
- Write your own code. Add comments that explain the queue/stack and the main steps.

The two example graphs



Graph 1 (Tasks 1 and 2), undirected, no weights.
Edges: A-B, A-C, B-D, B-E, C-F, E-F, F-G.



Graph 2 (Task 3), undirected, weights are minutes/cost.
Edges: A-B 4, A-C 2, B-C 1, B-D 5, C-D 8, C-E 10, D-E 2, D-F 6, E-F 3.

Neighbour lists (Graph 1)

Node	**A**	**B**	**C**	**D**	**E**	**F**	**G**
Neighbours	B, C	A, D, E	A, F	B	B, F	C, E, G	F

Points

Task	**Algorithm**	**Points**
1	BFS - shortest path (fewest edges)	30
2	DFS - iterative, with an explicit stack	30
3	Dijkstra - cheapest paths	40
	Total	100

Task 1 - Breadth-First Search (30 points)

Idea. BFS visits nodes level by level using a **queue** (first in, first out). Mark a node as discovered when you **put it in the queue**. Because BFS counts edges, the first time it reaches a node is along a path with the **fewest edges**; the *parent* array lets you rebuild that path.

Pseudocode

```
BFS(graph, start):
    create an empty QUEUE Q
    mark start as discovered;  dist[start] = 0;  parent[start] = none
    enqueue(Q, start)
    while Q is not empty:
        u = dequeue(Q)                      // take from the FRONT
        visit u                             // print it
        for each neighbour v of u (A to Z):
            if v is not discovered:
                mark v as discovered
                dist[v] = dist[u] + 1;  parent[v] = u
                enqueue(Q, v)               // add at the BACK
```

A. Hand trace on Graph 1, start A (8 points)

Step	Node dequeued	Newly enqueued	Queue after the step (front ... back)
init	-	A	A
1	A	B C	B C
2	B	D E	C D E
3	C	F	D E F
4	D	-	E F
5	E	-	F
6	F	G	G
7	G	-	empty

Visit order: A B C D E F G

B. Program (18 points)

Write a program that runs BFS on Graph 1 and prints the lines below. The shortest-path lines must be computed with *parent* and *dist* (rebuild the path from the target back to the start). Handle the case 'no path'.

```
BFS order from A: A B C D E F G
Shortest A -> G: distance 3, path A -> C -> F -> G
Shortest D -> G: distance 4, path D -> B -> E -> F -> G
```

C. Short answers (4 points)

- Why do we mark a node when it is **enqueued** and not when it is dequeued?
- What is the running time of BFS (V nodes, E edges)? Why does BFS fail to give cheapest paths on a weighted graph?

1. Marking on enqueue stops other neighbours from adding the same node again. It also keeps the first distance and parent found for that node.

2. O(V + E) with adjacency lists already in alphabetical order. Each node is queued once and each undirected edge is checked twice. BFS counts edges, so fewer edges can still have a larger total weight than another path.

Task 2 - Depth-First Search (30 points)

Idea. DFS goes as deep as possible along one branch, then **backtracks**. Use an explicit **stack** (last in, first out). A node is marked when it is **popped**, so the same node can be in the stack twice and the second copy is **skipped**. Pushing neighbours in **reverse** order makes the alphabetically first neighbour come out first.

Pseudocode

```
DFS(graph, start):
    create an empty STACK S
    push(S, start)
    while S is not empty:
        u = pop(S)                          // take from the TOP
        if u is already visited:  continue  // skip the second copy
        mark u as visited;  visit u         // print it
        for each neighbour v of u (Z down to A):     // REVERSE order
            if v is not visited:
                push(S, v)
```

A. Hand trace on Graph 1, start A (8 points)

Step	Node popped	Action	Pushed (in this order)	Stack after (top first)
init	-	push A	A	A
1	A	visit A	C B	B C
2	B	visit B	E D	D E C
3	D	visit D	-	E C
4	E	visit E	F	F C
5	F	visit F	G C	C G C
6	C	visit C	-	G C
7	G	visit G	-	C
8	C	skip	-	empty

Visit order: A B D E F C G

B. Program (18 points)

Write DFS with an **explicit stack** (no recursion) and print the lines below. Use the second start node G to prove that your program really works for any start.

```
DFS order from A: A B D E F C G
DFS order from G: G F C A B D E
```

C. Short answers (4 points)

- Why can a node be in the stack twice? In which step of your trace is a copy skipped?
- Compare the BFS order and the DFS order on Graph 1. Which data structure explains the difference?

1. Nodes are only marked on pop, so two neighbours can push the same node before it is visited. A pushes C, then F pushes it again. The older C is skipped at step 8.

2. BFS: A B C D E F G. Its FIFO queue visits nodes level by level.
   DFS: A B D E F C G. Its LIFO stack follows a branch before coming back.

Task 3 - Dijkstra's Algorithm (40 points)

Idea. Dijkstra finds the cheapest path from one source to every node when all weights are **non-negative**. Keep **dist** (best known cost, INF at the start, 0 for the source) and **prev** (node before it on that path). Repeatedly **settle** the unsettled node with the smallest dist, then **relax** its edges.

Pseudocode

```
Dijkstra(graph, source):
    dist[v] = INFINITY and prev[v] = none for every node v
    dist[source] = 0
    repeat while some unsettled node has dist < INFINITY:
        u = the unsettled node with the SMALLEST dist
        settle u                          // its dist is now final
        for each edge (u, v, w) where v is not settled:
            if dist[u] + w < dist[v]:       // relaxation
                dist[v] = dist[u] + w
                prev[v] = u
```

A. Hand trace on Graph 2, source A (10 points)

Step	Node settled (its dist)	Updates made (node = new dist via previous node)
init	-	A=0; B=C=D=E=F=INF; all prev=none
1	A (0)	B=4 via A; C=2 via A
2	C (2)	B=3 via C; D=10 via C; E=12 via C
3	B (3)	D=8 via B
4	D (8)	E=10 via D; F=14 via D
5	E (10)	F=13 via E
6	F (13)	no updates

Node	A	B	C	D	E	F
Final dist	0	3	2	8	10	13
Final prev	none	C	A	B	D	E

B. Program (24 points)

Implement Dijkstra on Graph 2 (a simple search for the minimum is enough; a heap is a bonus). Print one line after each settled node, then the cheapest path to every node.

```
init          | A=0 B=INF C=INF D=INF E=INF F=INF
settle A (0)  | A=0 B=4 C=2 D=INF E=INF F=INF
settle C (2)  | A=0 B=3 C=2 D=10 E=12 F=INF
settle B (3)  | A=0 B=3 C=2 D=8 E=12 F=INF
settle D (8)  | A=0 B=3 C=2 D=8 E=10 F=14
settle E (10) | A=0 B=3 C=2 D=8 E=10 F=13
settle F (13) | A=0 B=3 C=2 D=8 E=10 F=13
A to B: cost 3: A -> C -> B
A to C: cost 2: A -> C
A to D: cost 8: A -> C -> B -> D
A to E: cost 10: A -> C -> B -> D -> E
A to F: cost 13: A -> C -> B -> D -> E -> F
```

C. Short answers (6 points)

- Explain **relaxation** with one line of your table (for example, why B changes from 4 to 3).
- Why does Dijkstra give wrong results when an edge has a negative weight? How is BFS related to Dijkstra when every weight is 1?

1. Through C, B costs 2 + 1 = 3 instead of 4, so dist[B]=3 and prev[B]=C.
2. A negative edge may reduce a settled distance, so that distance may not be final.
   With all weights 1, BFS finds the same shortest distances using a FIFO queue.

Submission checklist

- One source file per task (or one file with three clearly marked parts) in the language of your choice.
- The output of your programs matches the expected output exactly.
- Hand-trace tables and short answers are completed (on this sheet or on separate pages).
- Comments explain the queue / stack and the relaxation step. Your name and ID are at the top of every file.
- Upload to the course page before the deadline given in class.

Hints for any language

- A **queue** needs two indexes (front, back) or a built-in deque. A **stack** needs one index (top) or a list with push/pop at the same end.
- Map the letters A, B, C, ... to the numbers 0, 1, 2, ... and keep an array of names for printing.
- Use a very large number as INFINITY, and a special value (for example -1) for 'no previous node'.
- Execute your algorithm by hand on paper first. Print the queue / stack inside the loop while debugging.
- Test with a second start node. A program that only prints the expected text for one start does not earn the points.